

How Police Funding Affects Asset Forfeitures: Evidence from a Regression Discontinuity Design

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Abstract

Asset forfeiture is a controversial policy that allows police departments to confiscate assets related to crimes and retain a portion of their value as revenue. This policy was explicitly intended to create a financial incentive for police to prioritize high-level drug crimes over minor offenses, and was successful in doing so. Providing a mechanism through which law enforcement can police for profit might introduce incentives beyond those intended. In this paper, I ask whether exogenous changes in police funding affect asset forfeitures. Specifically, do police departments respond to fluctuations in finances by adjusting the amount of assets they seize? Receiving funding from other sources could alleviate pressure for police agencies to collect revenue through asset forfeitures, causing seizures to decrease. Conversely, receiving extra resources may allow police departments to expand operating capacity and seize *more* assets. I employ a regression discontinuity design with data from a grant program where police departments apply for funding, are given a numerical score, and receive payouts based on whether their score is above a cutoff threshold, allowing for comparison between agencies that received funding and those that did not. I find that agencies that receive funding increase asset forfeitures relative to those that did not, pointing towards a mechanism where additional funding alleviates capacity constraints that limit the extent to which agencies can seize assets.

1 Introduction

1.1 Asset Forfeitures and Equitable Sharing

Civil asset forfeiture is a legal mechanism through which law enforcement agencies can permanently confiscate, either for sale or use, assets thought to be related to crime (Carpenter II, 2023 [1]). Asset forfeiture laws were originally enacted in the United States as a key component of the war on drugs (Boudreaux and Pritchard, 1996 [2]). Allowing police to profit from the confiscation of assets related to crimes was explicitly intended to incentivize law enforcement to prioritize high-level drug busts over petty nonviolent offenses, as these are the types of crimes most likely to be lucrative for the police agencies. A drug kingpin is far more likely to have cars, jewelry, and guns that the police can sell at auction than someone shoplifting.

Asset forfeiture laws varied by state (Edgeworth, 2008 [3]). In some states only drug-related crimes were eligible, while in others assets could be seized from any type of crime. In addition, there was variation in the percentage of revenue retained by the seizing department. In certain states nearly all of the funds went directly back to the police, while in others only a small portion did, with the rest being distributed to other government programs and agencies. Lastly, state laws varied as to how the money could be spent. In the most generous states there were essentially no restrictions, while in others money could only be spent on things like training or school safety programs.

The controversy behind asset forfeiture largely began in 1984, with the passage of the Comprehensive Crime Control Act, which created a program known as Equitable Sharing. Equitable Sharing created a federal channel through which seized assets could be processed, allowing law enforcement agencies to bypass their state procedures and regulations. Through equitable sharing, state and local law enforcement agencies could have cases "adopted" by the federal government. This meant that they could either seize the assets themselves and then turn them over to the federal government for processing, or have the federal government

adopt the case before the fact, and work in collaboration with a federal agency, such as the FBI or DEA, to seize the assets. Equitable sharing was far more generous than almost all state asset forfeiture programs. Local agencies received 80% of all revenues from assets processed through adoption, and the equitable sharing law required that there be no restrictions on how these funds could be spent (Kantor, Pawlowksi, and Kitchens 2021 [4]).

For these reasons, equitable sharing has been the target of much of the criticism towards asset forfeiture. There was major backlash in 2013 and 2014 following a number of high profile cases of abuse of the law (Stillman 2013 [5]), and large-scale reforms were implemented in 2015, causing a drastic reduction in usage of equitable sharing adoption. Most of these reforms, however, have been walked back in the years since, and asset forfeiture through equitable sharing is back on the menu (Stillman 2017 [6]).

Providing a mechanism through which law enforcement may police for profit may allow police behavior to be altered in ways beyond those intended. If funds from asset seizures are highly liquid, increasing forfeitures may be a simple way in which law enforcement agencies are able to increase revenues, particularly when funding from other sources is lost. Although the original goal of incentivizing police to prioritize high-level drug crimes over minor offenses may be desired, it could still be sub-optimal for forfeiture behavior to fluctuate with other factors such as police funding. To ensure that law enforcement is fair and effective, it is important to understand the full range of incentives driving police behavior.

1.2 Related Literature

Different components of the incentives surrounding asset forfeiture have been studied before. Baicker and Jacobson (2007 [7]) look at the effectiveness of the intended incentives created by asset forfeiture laws, as well as the local government response. They find that police in states with more "generous" forfeiture laws (in terms of the share of proceeds police may keep) increased seizures by more than in places with less generous forfeiture laws. They additionally look at the local government response through police budget allocations. Their

results show that when a police agency increases revenue through forfeitures, local governments offset this gain somewhat by reducing the funding they are allocated in the annual budget. This offset is not 1:1 however, so the agency still profits from increasing forfeitures. This "profit net of budget reductions" is what police behavior responds to, not the explicit sharing provisions laid out in the law.

Holcomb, Kovandzic, and Williams (2011 [8]) analyze the relationship between state forfeiture provisions and the federal equitable sharing program. They find that in states where forfeiture laws are more restrictive, police agencies process a larger share of seizures through the federal program, essentially using federal equitable sharing to circumvent state restrictions.

Kantor, Pawlowksi, and Kitchens (2021 [4]) examine how incentivizing law enforcement through asset forfeiture affects crime outcomes. They find that when equitable sharing policies were enacted, law enforcement responded to the incentive as intended, shifting efforts from other crimes to drug enforcement. This reduced crime overall, particularly in places where the new equitable sharing provisions were more generous than the previous state provisions, thus creating a stronger incentive. This change did, however, lead to an uptick in traffic fatalities, likely as a result of weaker enforcement of road laws as drug crimes were prioritized.

Worall (2001 [9]) documents the extent to which forfeiture revenues have become a significant portion of many police budgets, which he describes as an "addiction." In a survey, nearly 40% of responding agencies said that "civil forfeiture is necessary as a budgetary supplement." This result squares nicely with what is known of local government budget responses from Baicker and Jacobson (2007 [7]). As police increase revenues from seizures, budget allocations are reduced, and so they must again increase revenues from seizures in a vicious cycle.

Makowsky, Stratmann, and Tabarrok (2019 [10]) come closest to sharing a research question with this paper. They examine how forfeiture behavior changes in areas experienc-

ing budget deficits, and find that arrests and property seizures increase with local deficits. Their paper is somewhat limited in scope in that data for their analysis is available only for 2007 and 2012, their deficit/surplus measure is at the level of the entire local government rather than capturing funds available to the police department specifically, and collects forfeitures data through state programs only, potentially missing large portions of forfeitures that are processed through the federal equitable sharing program. Each of these components is something that this paper attempts to address.

1.3 COPS-CHP Grant

Relating asset forfeiture to police funding is not as simple as regressing assets seized on police budget allocations, for obvious reasons. In places where crime is higher, police may receive more funding and also have to seize more assets. Finding truly exogenous variation in funding that is not related to forfeitures is the primary challenge.

The Cops Hiring Program (CHP) is a grant distributed by the Department of Justice's Community Oriented Policing Services (COPS) office. This program began in 1994 and provides law enforcement agencies with money for hiring additional police officers. The program exceeded \$1 billion a year in the 90s, but is closer to \$200 million a year today. Law enforcement agencies apply for the grant by submitting statistics such as their current funding level, agency size, and local crime rates and trends, as well as written answers about long term goals and how the funds will be spent. Applications are given a crime score and a fiscal needs score by the COPS office, which are aggregated to an overall score. Applications are then funded in descending order until money for the year has been exhausted. This creates a cutoff score that varies each year, above which applicants receive funding and below which they do not. A minor caveat is that 1% of funds are required to go to each state, allowing certain agencies with application scores below the cutoff to receive the grant regardless, as long as they receive one of the highest scores in their state. This setup naturally lends itself to a fuzzy regression discontinuity design. Agencies directly on either

side of the cutoff will, on average, be similar along most margins, though some will have found themselves just above the threshold and receive extra funding and some will not. Which side of the cutoff each agency finds themselves is essentially random, and gives us our exogenous variation in funding. A small body of research such as Evans and Owens (2007 [11]) and Mello (2019 [12]) has used the COPS program for identification before, albeit to answer different questions. Cook, Kapustin, Ludwig, and Miller’s (2017 [13]) paper is arguably the most similar to this one in that it applies a regression discontinuity design to the COPS data.

2 Data

2.1 COPS Grant

Data on the COPS-CHP application scores was received through a FOIA request from the Department of Justice. The data is from 2010-2024, although the program was not run in 2017 and 2018. The data is at the agency-year level and contains the application scores of approved and denied applicants and the amount of funding received by those approved. Crucially, seven-digit ORI identifier codes used to identify police agencies and link data sets were only provided for the 2010-2020 period, excluding the 2021-2024 data from the main analysis. Funding amounts were also excluded from the 2021-2024 data. See Table 1 for summary statistics of the annual application scores and Figure 1 for a visual representation.

Table 1: COPS-CHP Grant Application Score Summary Statistics

Year	Mean	St. Dev.	Cutoff	% Funded	N
2010	59.27	13.04	80.65	8.24	4,402
2011	134.78	21.10	170.25	8.31	2,701
2012	29.17	6.37	38.015	15.59	1,052
2013	110.25	17.00	131.68	15.29	1,714
2014	108.09	18.61	128.63	16.27	1,293
2015	109.59	19.68	129.66	19.07	1,095
2016	109.17	20.55	132.87	15.52	1,177
2017	123.98	25.38	157.27	15.67	1,117
2020	112.05	24.41	117.58	54.59	1,090

Note: Data from COPS office of the Department of Justice

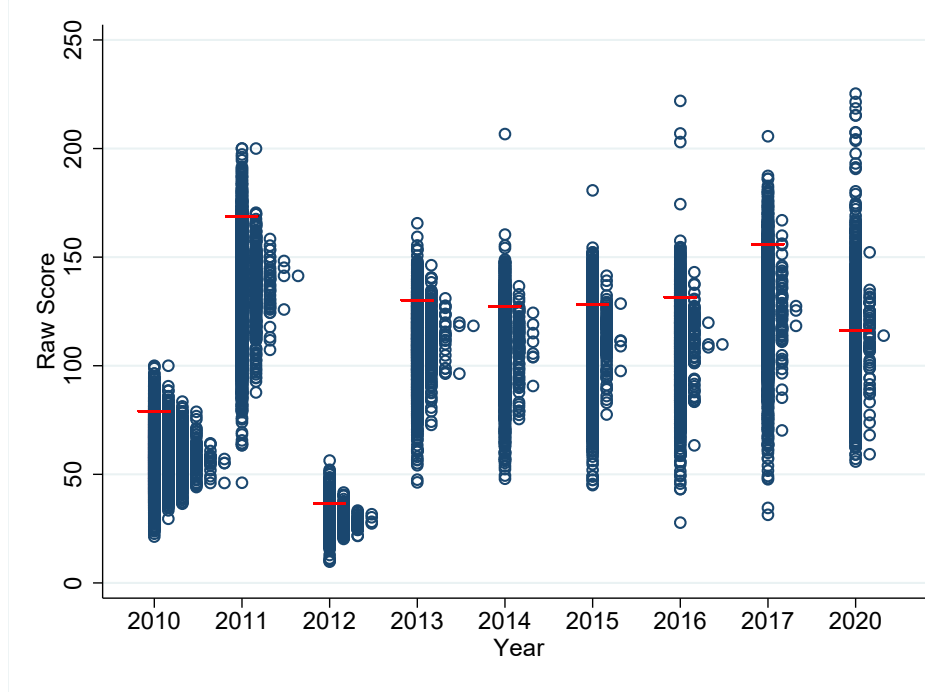


Figure 1: Annual COPS-CHP Grant Application Scores

Note: Red lines display funding score cutoff for each year

CHP funding surged to just shy of \$1 billion in 2009 when President Obama signed the American Recovery and Reinvestment Act. Funding levels dropped significantly the following year to the neighborhood of \$300 million. Figure 2 shows CHP funding amounts for our entire sample period (and 2009). Table 2 presents statistics on the dollar amounts of

grants awarded each year in the sample period.

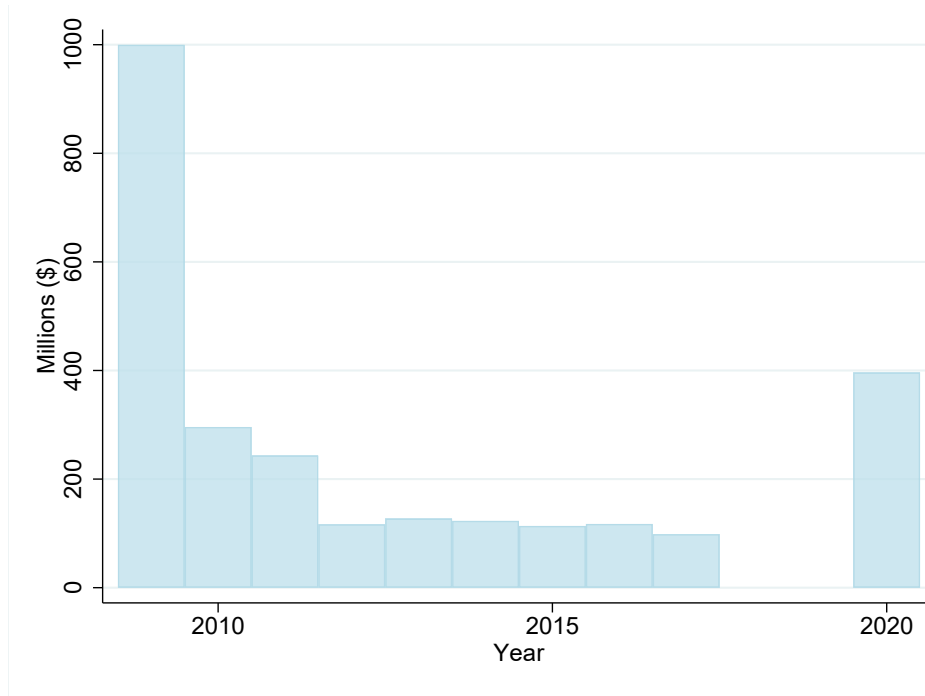


Figure 2: CHP Funding 2009-2010

Table 2: Annual CHP Awards

Year	Mean	Median	Max	N
2010	\$785,139	\$239,331	\$21,382,750	377
2011	\$1,022,684	\$387,719	\$11,306,450	238
2012	\$500,583	\$157,840	\$6,428,350	233
2013	\$478,384	\$250,000	\$4,515,730	266
2014	\$575,655	\$250,000	\$3,634,605	214
2015	\$544,227	\$250,000	\$3,125,000	209
2016	\$641,718	\$352,879	\$3,513,921	183
2017	\$550,254	\$250,000	\$3,125,000	179
2020	\$654,887	\$250,000	\$11,563,600	605

2.2 Asset Forfeitures

Data on seized assets comes from two sources. The Consolidated Asset Tracking System (CATS), housed by the Department of Justice, is a database of all assets seized through the federal equitable sharing program, from 1996 through 2024. This data is on the asset

level, and contains information such as dollar value, date of seizure, and which local agency the asset was seized by, identified by seven-digit ORI code. The Equitable Sharing Agreement and Certification (ESAC) dataset, also housed by the Department of Justice, tracks the payments dispersed to police agencies for seized assets processed through the equitable sharing program. This data is at the agency-year level, and contains the dollar amount the agency received in equitable sharing payments as well as total operating budgets reported by the agency. Agencies are again identified via seven-digit ORI code. The ESAC data is only available, however, from 2016 through 2024, meaning it overlaps with the COPS data that contains ORI codes for only three years: 2016, 2017, and 2020. For this reason, the CATS data is the primary source that will be used for the main analysis. Additionally, there can be significant time lag between when an asset is seized and when it is fully processed and payment is dispersed back to the seizing agency, so payments in the ESAC data do not accurately reflect levels of forfeitures seized in that year. Asset forfeitures (from the CATS data) reached all-time highs in the early 2010s before falling due to rising criticisms and temporary policy reform. Figure 3 shows the evolution of the forfeiture levels before and during our sample period. Figure 4 shows the breakdown of seized assets by asset type, the vast majority being cash.

Table 3 displays total operating budgets as reported in the ESAC data, seizure amounts from the CATS data, and re-displays the COPS funding amounts. Seizures make up a relatively small share of operating budgets, in the neighborhood of 1-2%, with COPS funding around an order of magnitude greater. These funding levels are relevant: body worn camera purchases, IT software licenses, and training programs in areas such as mental health crisis response or de-escalation are each components that represent 1-2% of budgets for small police departments (Strom 2017 [14])

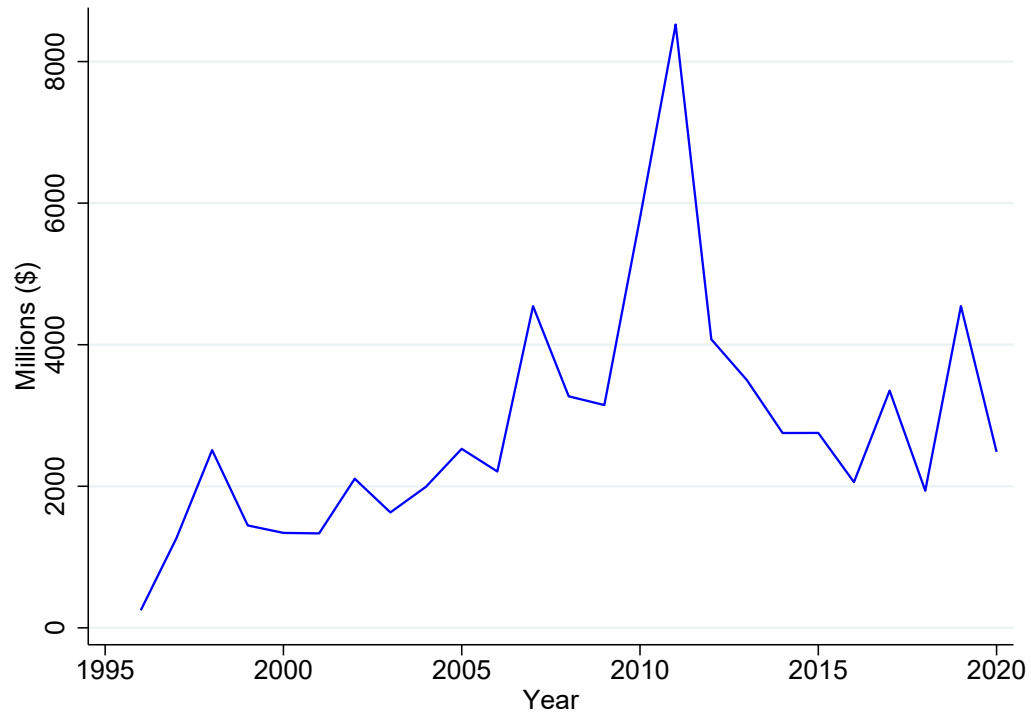


Figure 3: Annual Asset Forfeitures Through Equitable Sharing Adoption

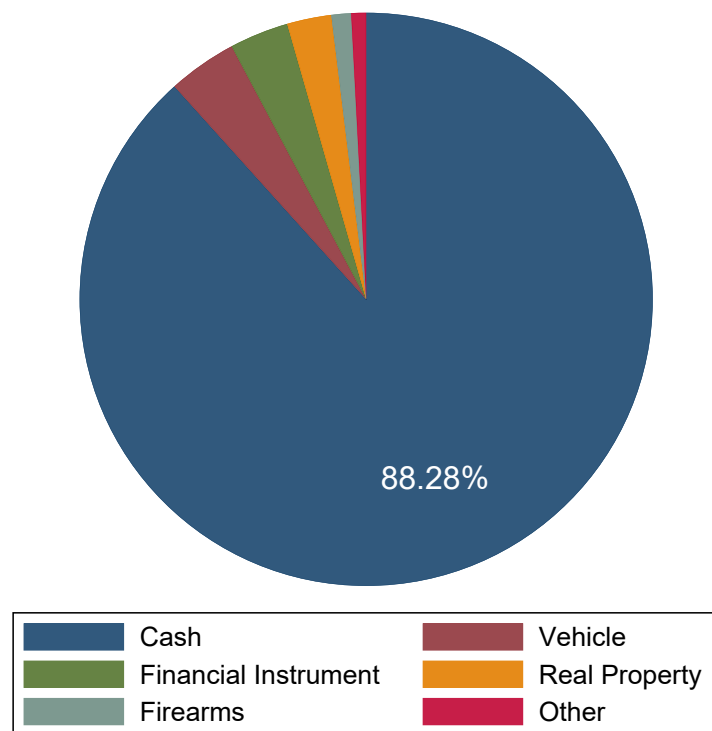


Figure 4: Seized Asset Categories, 1996-2020

Table 3: Summary Statistics of Budget, Seizures, and COPS Funding

Year	Budget		Seizures		COPS Funding	
	Mean	Median	Mean	Median	Mean	Median
2016	\$22,471,987	\$3,795,920	\$105,068	\$21,927	\$641,718	\$352,879
2017	\$51,455,216	\$4,561,117	\$134,482	\$27,005	\$550,254	\$250,000
2018	\$28,421,703	\$4,878,601	\$82,831	\$21,900	-	-
2019	\$33,630,712	\$5,145,686	\$283,824	\$21,433	-	-
2020	\$40,155,330	\$5,388,479	\$94,804	\$21,893	\$654,887	\$250,000

2.3 Supplemental Data

FIPS-level population data comes from the US Census Bureau. Data on unemployment rates, also at the FIPS-level, comes from the U.S. Bureau of Labor Statistics. Agency-level data on reported crimes as well as arrests comes from the FBI’s Uniform Crime Reporting Program. Agency-level personnel counts of both officers and civilian employees come from the Law Enforcement Officers Killed in Action (LEOKA), also from the FBI.

3 Research Design

3.1 Sample

The asset-level CATS data was collapsed to the agency-year level, giving an annual dollar amount of seized assets for each reporting agency. This was linked to the COPS application data using ORI identifier. In the sample period of 2010-2020 (excluding 2018 and 2019), 3,929 agency-year pairs were successfully merged between these two data sets. If an agency did not appear in the CATS data in a given year, rather than dropping it from the sample, they are treated as having 0 seized assets for that year, as agencies are only required to submit the ESAC compliance form and thus appear in the data if they are to receive equitable sharing payments (Department of Justice [15]). This brings us to a final sample of 15,641 agency-year observations. Summary statistics of selected descriptive variables of the sample

are presented in Table 4.

Table 4: Summary Statistics Below and Above the Cutoff

	Below Cutoff			Above Cutoff		
	Mean	St. Dev.	N	Mean	St. Dev.	N
Property Crimes Per 10k	102.34	157.55	12,472	153.63	154.40	1,420
Violent Crimes Per 10k	46.53	68.59	12,472	83.62	96.32	1,420
Officers Per 10k	8.78	18.38	12,474	9.85	11.31	1,420
FIPS Unemployment Rate	7.92	2.82	12,633	8.11	3.22	1,422

A large share, just over 75% of observations, are associated with zero forfeitures. Table 5 provides a more detailed breakdown of the distribution of asset forfeiture funds earned over the period. In 2011, Google agreed to forfeit \$500 million for "allowing Canadian pharmacies to place advertisements through Google's AdWords program" (National Guard [16]). This resulted in two Rhode Island Police departments, East and North Providence, to receive massive payouts of \$60 million each, as their counter-drug units were involved in the investigation. For this reason, the maximum amount excluding these two agencies is also included in Table 5.

Table 5: Asset Forfeitures Percentile Distribution

Percentile	Asset Forfeitures (\$)
Min	0
10th	0
25th	0
50th	0
75th	0
80th	5,288
90th	55,073
99th	770,330
3rd Max	8,119,764
Max	60,238,896

Because the sample includes multiple years pooled together, and the distribution and

cutoff of the scores can change greatly from year to year as seen in Table 1, to compare applicants across years scores are standardized by subtracting that year’s cutoff score and dividing by that year’s standard deviation:

$$StandardScore_{ti} = \frac{(RawScore_{ti} - Cutoff_t)}{StDev_t} \quad (1)$$

This leaves us with standardized values that represent how many standard deviations a score is above or below the cutoff for that year.

Given the wide range of police department sizes and capacities, it is best to think about changes in asset forfeiture in terms of percentages rather than levels. Calculating the percent change in asset forfeitures is somewhat complicated by the prevalence of zero values in the data. If a police agency goes from zero forfeitures in 2015 to \$10,000 of forfeitures in 2016, their percent change, as usually calculated, is undefined. Percent changes are thus approximated using the inverse hyperbolic sinh: $\text{arcsinh}(y) = \ln(y + \sqrt{y^2 + 1})$. Bellemare and Wichman (2020 [17]) note potential concerns when applying the inverse hyperbolic sinh, as for large values of y it approximates $\ln(2y)$. Thus, for robustness, results calculating percent changes using $\ln(y + 1)$ and the midpoint rule will be reported as well.

3.2 Fuzzy RD

As previously mentioned, the most applicable design for this setting is a fuzzy regression discontinuity, as the probability of receiving funding jumps discontinuously at the cutoff. The equation to estimate is:

$$Y_{it} = \beta_0 + \beta_1 * D_{it} + f(S_{it} - 0) + \delta_t + \alpha_s + \varepsilon_{it} \quad (2)$$

where Y_{it} is the outcome of interest (percent change in asset forfeitures for agency i in year t), D_{it} is agency i ’s treatment status (funded versus unfunded) in year t , S_{it} is agency i ’s application score in year t , δ_t are year fixed effects to control for time-varying heterogeneity

as this is a pooled sample comprising many years, α_s are state fixed effects to control for state-level differences in policy (police funding allocations, regulations on usage of funds, etc.), and ε_{it} is the idiosyncratic error term.

In practice, fuzzy RD is estimated in two stages. The first stage estimates the likelihood of treatment given a score and its distance from the cutoff. The estimating equation for the first stage is:

$$D_{it} = \beta_{0'} + \beta_{1'} * Z_{it} + f(S_{it} - 0) + \delta_t + \alpha_s + \varepsilon_{it} \quad (3)$$

where Z_{it} is an indicator for agency i 's score in year t being above the year t cutoff. The second stage then estimates the outcome given a predicted treatment status, $\hat{D}_{it}$, from the first stage:

$$Y_{it} = \beta_0 + \beta_1 * \hat{D}_{it} + f(S_{it} - 0) + \delta_t + \alpha_s + \varepsilon_{it} \quad (4)$$

The coefficient β_1 from equation (4) can then be interpreted as an estimate of the intention-to-treat (ITT). An identical way to think about fuzzy RD is to think about the local average treatment effect (LATE) as the ratio of the jump in outcome over the jump in treatment probability at the cutoff, a local Wald estimator. Explicitly:

$$\widehat{\text{LATE}} = \frac{\widehat{E}[Y \mid X = c^+] - \widehat{E}[Y \mid X = c^-]}{\widehat{E}[D \mid X = c^+] - \widehat{E}[D \mid X = c^-]}. \quad (5)$$

where as before Y is our outcome variable, D is a binary for treatment status, and $X = c^+$ and $X = c^-$ denote observations above and below the cutoff, respectively.

Following the likes of Porter (2003 [18]) and Calonico, Cattaneo, Farrell, and Titiunik (2019 [19]), I estimate the fuzzy regression discontinuity non-parametrically using a local polynomial regression around the cutoff, avoiding imposing a functional form. Following Calonico, Cattaneo, and Farrell (2020 [20]), I select Coverage Error Rate (CER) optimal bandwidths separately for either side of the cutoff. Contrary to mean squared error (MSE) optimal bandwidths, which minimize error in the point estimate, CER optimal bandwidths minimize error in confidence interval coverage and are thus better for valid inference.

3.3 Assumptions

Two primary assumptions must hold for the validity of any regression discontinuity design. The first is that there is no precise manipulation of the running variable near the cutoff. In this setting, this means that it must be impossible for any police agency to determine their own score with enough precision as to place themselves directly on either side of the cutoff. Of course, each applicant will be trying to maximize the strength of their application (and thus their score) in order to maximize their chances of receiving funding. This is not a cause for concern as it simply means each agency is trying to place themselves as far to right of the distribution as possible, rather than directly to the right of the cutoff. McCrary (2008 [21]) proposes a test for bunching at the cutoff that has become standard in the regression discontinuity literature. The McCrary test compares densities immediately to either side of a cutoff c by estimating a log difference in heights:

$$\hat{\theta} = \log \left(\frac{f_-(c)}{f_+(c)} \right) \quad (6)$$

where $f_+(c)$ and $f_-(c)$ are the estimated densities to the right and left of the cutoff, and testing the hypothesis that the density is continuous ($\hat{\theta} = 0$). If the result of the test is significant, there is evidence of sorting. I find a p-value of 0.42, an insignificant result, meaning there is no evidence of sorting. Figure 5 shows this graphically, raising no visual suspicion.

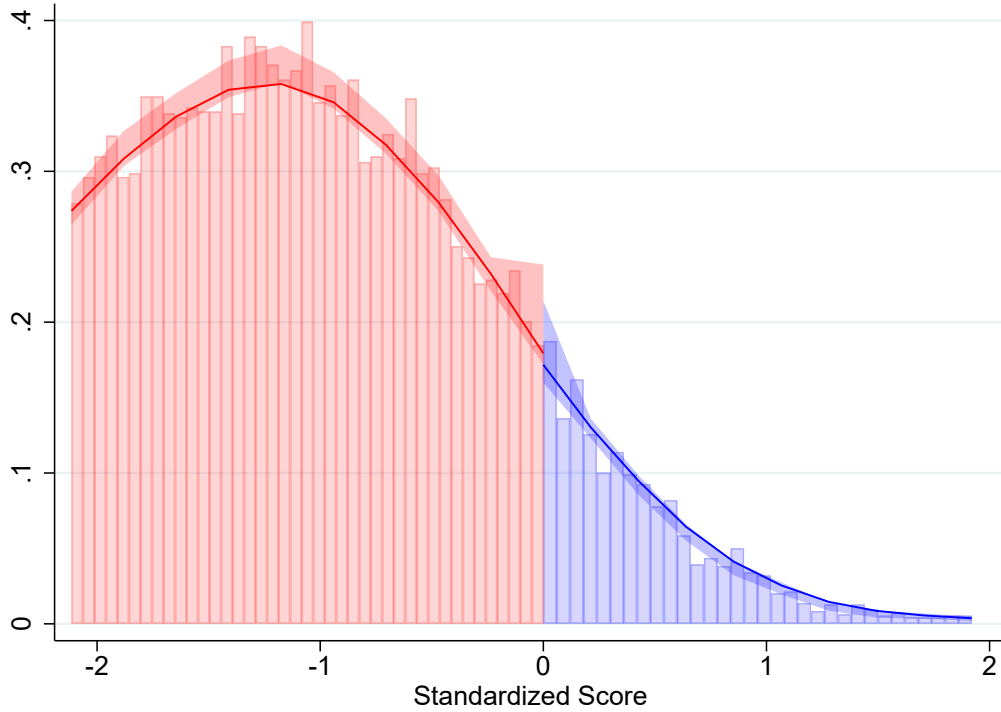


Figure 5: Application Score McCrary Test

Another key assumption for validity of the regression discontinuity is that other relevant variables are continuous at the cutoff threshold. If other measures that may have a causal effect on the outcome also jump discontinuously at the cutoff it would be impossible to attribute any estimated effects to the treatment. To examine this I selected four variables from the available data that may be likely candidates of measures that are both related to the COPS application scoring process (and thus have a possibility of jumping discretely at the cutoff) and also might be correlated with the outcome measure: officers per 10,000 persons, violent crimes per 10,000 persons, property crimes per 10,000 persons, and unemployment rate. Figure 6 displays how these measures evolve at the cutoff, estimating local polynomials on either side. A small discontinuity can be seen only for property crimes per 10,000. Table 6 reports the results of regression discontinuities estimated with each of these variables as the outcome. In agreement with what is seen visually, property crimes per 10,000 is the only statistically significant coefficient, though only at the 5% level. While this potential confounder

is certainly something to keep in mind when interpreting future results, the magnitude of the discontinuity is relatively small and the coefficient is only marginally significant, limiting the extent to which it causes concern.

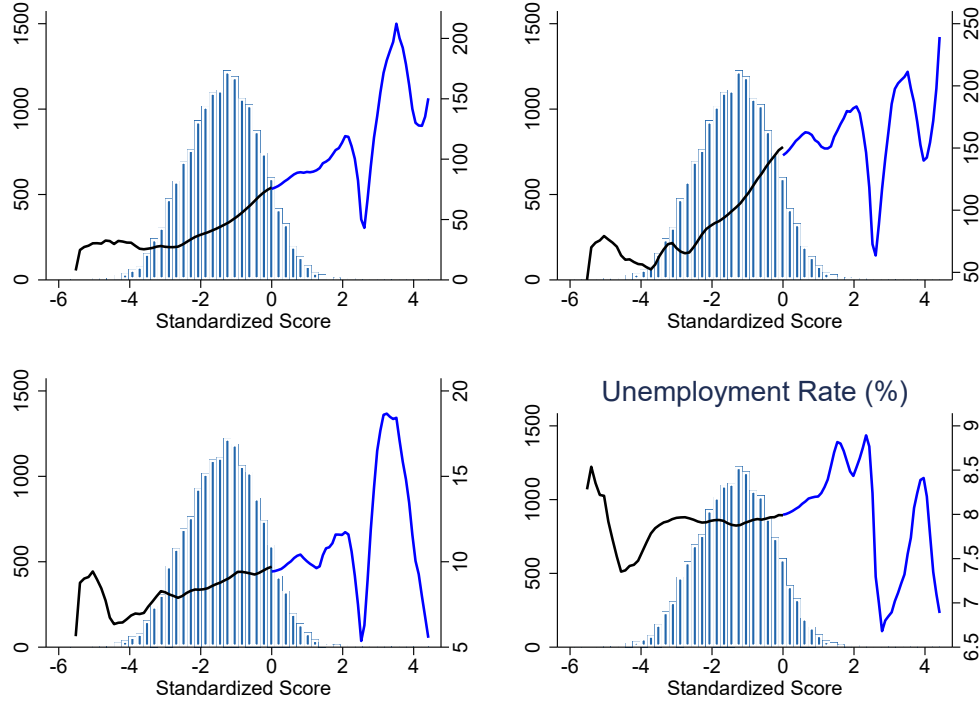


Figure 6: Continuity Check

Table 6: Continuity Check: Regression Discontinuity Estimates

	(1) Officers Per 10k	(2) Violent Crimes Per 10k	(3) Property Crimes Per 10k	(4) Unemployment Rate
	-0.738 (0.799)	-11.422 (6.753)	-25.161* (11.689)	-0.048 (0.160)
Year FE	Yes	Yes	Yes	Yes
P-Value	0.356	0.091	0.031	0.763
N	13,894	13,892	13,894	14,055
Kernel	Triangular	Triangular	Triangular	Triangular

Notes: Robust standard errors are shown in parentheses. * indicates significance at the 5% level.

4 Results

Table 7 below presents the estimates of Equation (2). As mentioned before, results are presented with the percent change in asset seizures calculated using both inverse hyperbolic sinh (columns (1)-(3)) and $\ln(y + 1)$ (columns (4)-(6)). As the table shows, the results are quantitatively similar regardless of the method used. The percent changes that are represented by the coefficients are marginally smaller in the specifications using logs, in line with what is to be expected if $\operatorname{arcsinh}$ is beginning to approximate $\ln(2y)$ for larger values. Columns (2), (5), and (8) present results including year fixed effects. As our sample is pooled from many years, year fixed effects control for time-varying heterogeneity from both the COPS side, such as changes in grant funding amounts year to year (see Figure 2), and the seizures side, such as changes in policy or culture that may affect forfeitures (see Figure 3). Columns (3), (6), and (9) include state fixed effects as well, controlling for heterogeneity at the state level that may impact forfeitures. As Holcomb, Kovandzic, and Williams (2011 [8]) show, state forfeiture laws dictate the extent to which police agencies process forfeitures through the federal equitable sharing program as opposed to the state programs. As such, there may be a systematic relationship between the state in which a police agency resides and what share of their forfeitures appear in our federal data. Reassuringly, results are quantitatively similar between all fixed effects specifications.

A coefficient of about 0.95 means that for an agency near the threshold, being randomly placed above the cutoff and receiving CHP grant funding is associated with a 95% increase in the amount of assets seized relative to an agency that finds themselves just below. This number might seem incredibly large. Much of this is due to the way both $\ln(y + 1)$ and the inverse hyperbolic sinh are calculated. Consider a scenario where in year 1, a police agency seized \$100,000 of assets, and the following year seized none. Calculating their percent change in assets using inverse hyperbolic sinh gives -12.260, and using $\ln(y + 1)$ gives -11.513, an over 1,000% decrease with either method. In reality, going from any number to 0 should be thought of as a 100% decrease. Consider another example where a police agency

goes from zero seizures in year 1 to \$5,000 in year 2. Calculating their percent change in assets using inverse hyperbolic sinh gives 9.21, and using $\ln(y + 1)$ gives 8.517, both nearly 1,000% percent increases for a relatively small increase in seizures. It is trickier to think about the "true" percent change when an agency goes from zero forfeitures in year 1 to some amount of forfeitures in year 2. All of these instances could be thought of as 100% increases, but this would treat a jump from zero to \$100 the same as a jump from zero to \$1,000,000. Regardless, what is clear is that both arcsinh and $\ln(y + 1)$ calculate percent changes of an extremely large magnitude when the value in one of the periods is zero, a scenario that occurs in over 10% of the data and over 35% of the data excluding the cases where both values are zero.

To get some sense of the true scale of this effect, columns (7)-(9) report results calculating percent changes using the midpoint rule, which averages the values from years 1 and 2 as the denominator in a standard percent change calculation:

$$\% \Delta Y = \frac{Y_2 - Y_1}{(Y_1 + Y_2)/2} \quad (7)$$

While technically still undefined in cases where both values are zero, these instances are easily treated as a zero percent change. To some extent the midpoint rule still marginally overestimates the magnitude of certain percentages: going from any value in year 1 to zero in year 2 is calculated as a 200% decrease when it really should be thought of as a 100% decrease. Conversely, going from zero forfeitures in year 1 to some amount in year 2 is calculated as a 200% increase. Nonetheless, these bounds control magnitudes from exploding in a way that arcsinh and $\ln(y + 1)$ do not, without sacrificing much accuracy when values are of similar magnitudes. The statistical significance of these estimates is more marginal, though the specification with full state and year fixed effects is still significant at the 5% level. Coefficients around 0.250 represent that for an agency near the threshold, being randomly placed above the cutoff is associated with a 25% increase in the amount of assets seized relative to an agency that finds themselves just below. Figure 7 plots the regression

discontinuity result visually. The jump at the cutoff is somewhat noisy, as expected from the results in Table 7, but can be seen.

Two additional specifications attempt to get a better sense of the magnitudes of the effect, something that is muddled in the previous results by the zero values in the data. Table 8 columns (1)-(3) report results with the raw change in dollar amount as the outcome variable. Coefficients in the 5,000-7,000 range seem completely reasonable: an agency that is awarded COPS grant funding would increase the amount of assets they seize by less than \$10,000. Statistical power takes a considerable hit: no results are significant and all p-values are quite large. Nevertheless, this specification is the best bet at placing an explicit dollar amount to the effect of COPS funding on asset forfeitures.

Columns (4)-(6) return to log % change as the outcome variable, but with a restricted sample. In these specifications, only observations with non-zero values for both years are included. The sample size shrinks drastically, reducing statistical power, but the smaller coefficients are still informative as they put a loose range on the effect that one might expect to see amongst agencies that are always engaging in some level of asset forfeiture regardless. In these specifications, the effect of receiving COPS funding on asset forfeitures ranges from 25-35%, not too far off from the specifications using the midpoint method in Table 7, results that did achieve marginal significance.

Table 7: Primary Regression Discontinuity Estimates

	Inverse Hyperbolic Sinh			Log Change			Midpoint Rule		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	1.003*	1.008*	1.140*	0.950*	0.956*	1.078*	0.238	0.237	0.250*
	(0.456)	(0.453)	(0.467)	(0.428)	(0.425)	(0.439)	(0.124)	(0.123)	(0.126)
Year FE		Yes	Yes		Yes	Yes		Yes	Yes
State FE			Yes			Yes			Yes
BW (left)	0.597	0.613	0.542	0.595	0.611	0.540	0.582	0.617	0.545
BW (right)	0.373	0.377	0.337	0.370	0.374	0.333	0.324	0.323	0.290
Kernel	Triang.	Triang.	Triang.	Triang.	Triang.	Triang.	Triang.	Triang.	Triang.
P-Value	0.027	0.026	0.013	0.025	0.024	0.013	0.056	0.054	0.047
N	15,641	15,641	15,641	15,641	15,641	15,641	15,641	15,641	15,641

Notes: Robust standard errors are shown in parentheses. * indicates significance at the 5% level.

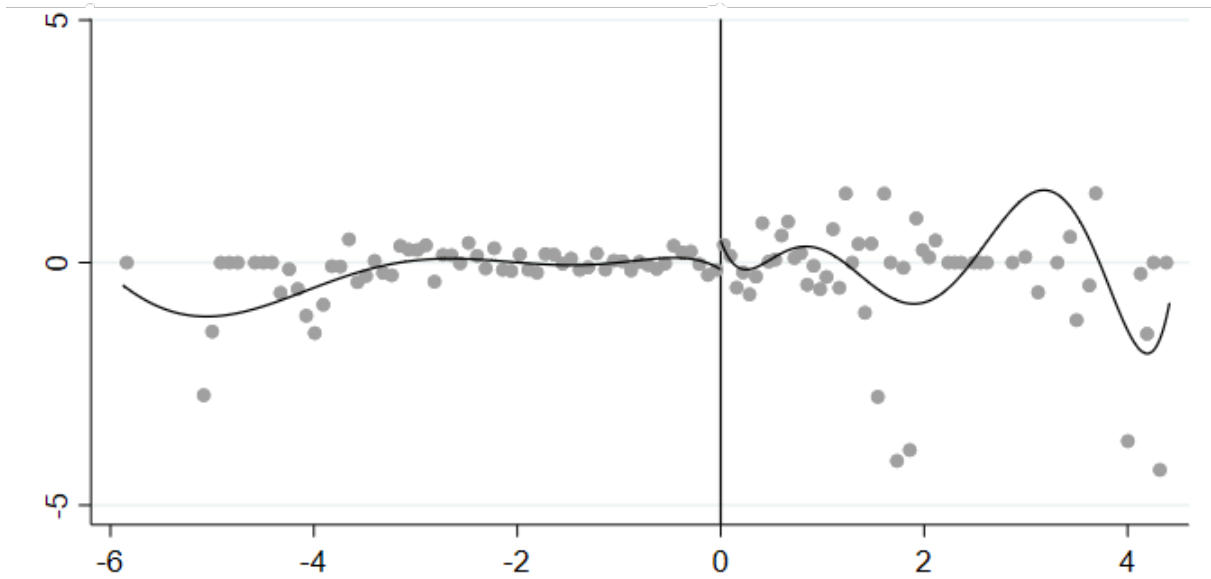


Figure 7: Arcsinh Regression Discontinuity Plot

Table 8: Regression Discontinuity Estimates: Raw Dollar Change and Restricted Sample

	Raw \$ Change			Log % Change: Non-zeros Only		
	(1)	(2)	(3)	(4)	(5)	(6)
	6,540 (20,277)	5,303 (19,887)	7,794 (19,336)	0.263 (0.414)	0.335 (0.411)	0.310 (0.412)
Year FE		Yes	Yes		Yes	Yes
State FE			Yes			Yes
BW (left)	0.320	0.327	0.319	0.643	0.705	0.574
BW (right)	0.352	0.371	0.391	0.345	0.320	0.313
Kernel	Triang.	Triang.	Triang.	Triang.	Triang.	Triang.
P-Value	0.747	0.790	0.687	0.525	0.416	0.451
N	15,641	15,641	15,641	2,922	2,922	2,922

Note: Standard errors are reported in parentheses.

5 Conclusion

The results show that when a police agency receives additional funding through the COPS-CHP grant program, the amount of assets they seize relative to agencies that did not receive funding increases. A potential explanation for this result is that in general police agencies are capacity constrained and receiving additional resources expands their ability to conduct law enforcement activity, including asset forfeitures. This explanation would square with previous results in the literature, such as Mello (2019 [12]), that receiving COPS funding reduces local crime rates. This is not necessarily to say that incentives are unrelated to the connection between funding and forfeitures. Asset forfeitures are just one item in a basket of activities that law enforcement agencies must delegate their time and resources across. It may be that when operating capacities are expanded, increasing asset forfeitures receives a higher priority than increasing other, less lucrative, law enforcement activities such as wellness checks. Exploring this possibility is a natural next step towards understanding the relationship between police funding and asset forfeitures.

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